Machine Learning

Diffusion Models

Dong-Kyu Chae

PI of the Data Intelligence Lab @HYU
Department of Computer Science
Hanyang University

Data
Intelligence
LAB

Diffusion Model Intro

DALLE·3
+
ChatGPT

You

기계학습이론을 듣는 한양대학교 학생들을 그려줘.

ChatGPT

Here is the image of Hanyang University students in South Korea attending a machine learning theory class.

Diffusion Model Intro

DALLE·3
+
ChatGPT

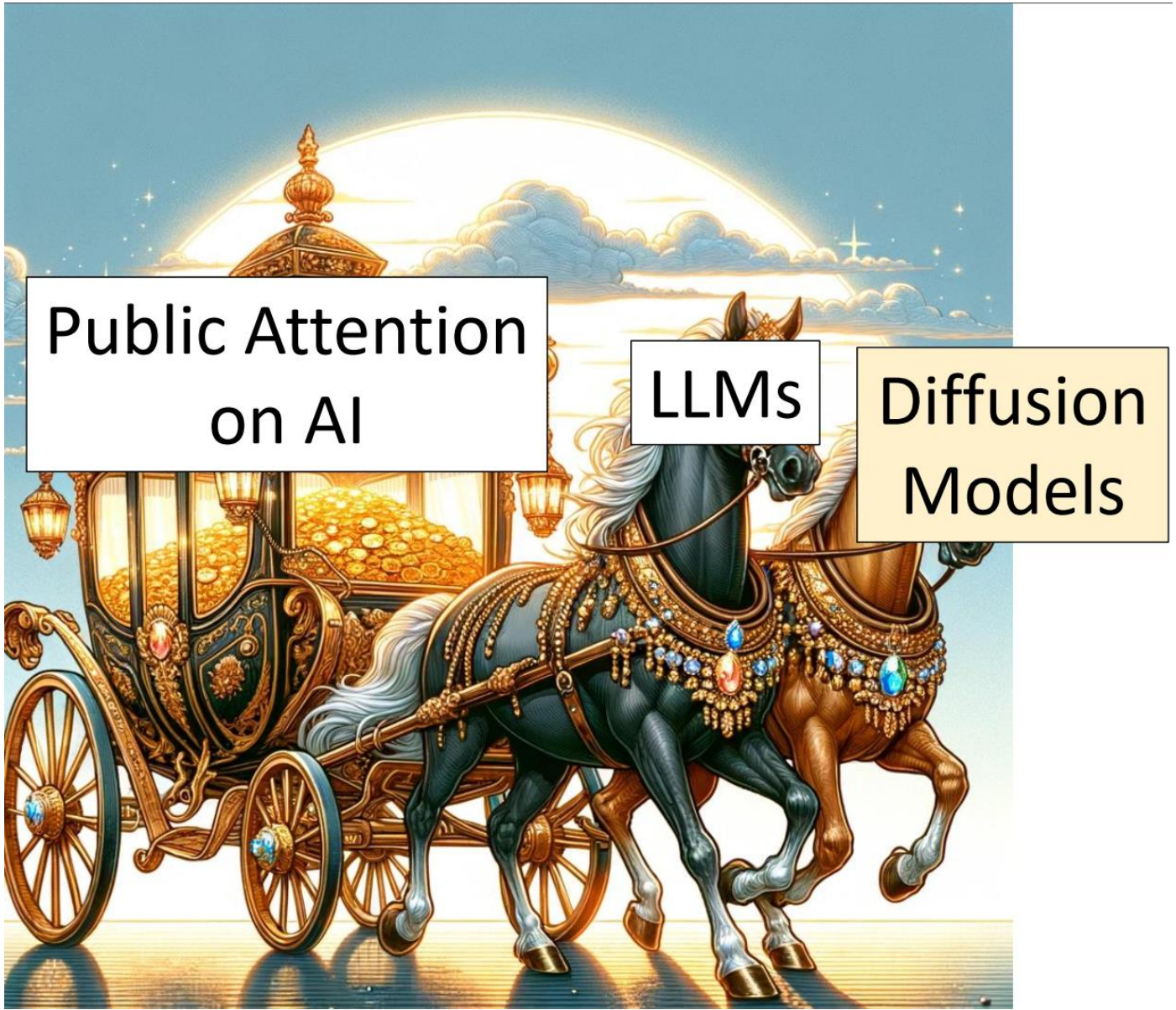
You

컴소인의 밤을 그려줘

ChatGPT

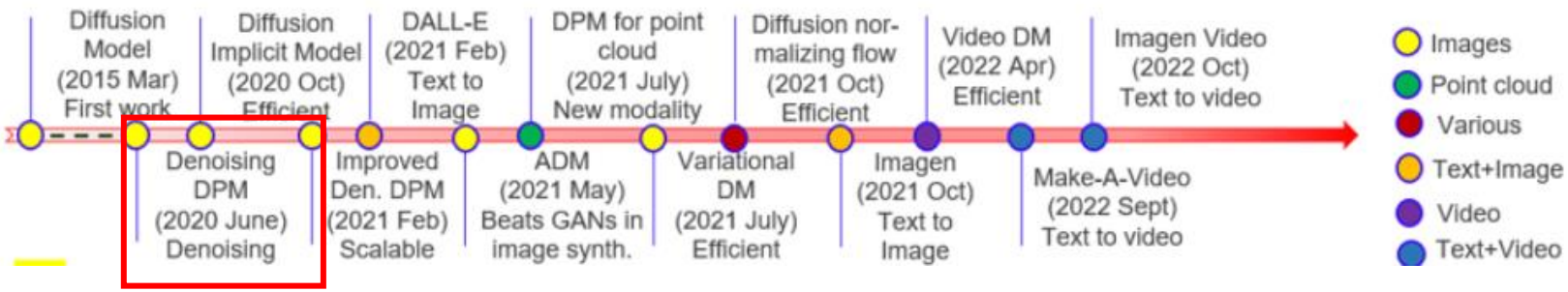
Here is the image depicting "Computer Science Night," a festive event for computer science students.

Diffusion Model Intro



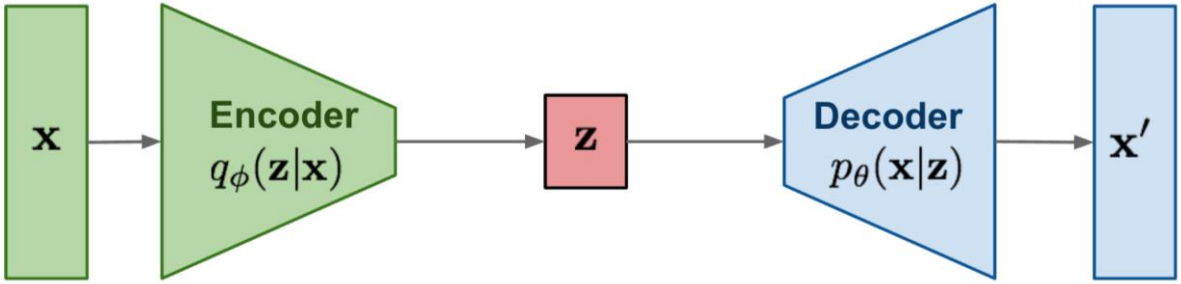
Diffusion Model Intro

- **What we will learn today**
  - https://arxiv.org/pdf/2006.11239.pdf Jonathan Ho, Ajay Jain, and Pieter Abbeel (UC Berkeley), "**Denoising Diffusion Probabilistic Models**" in NeurIPS 2020.

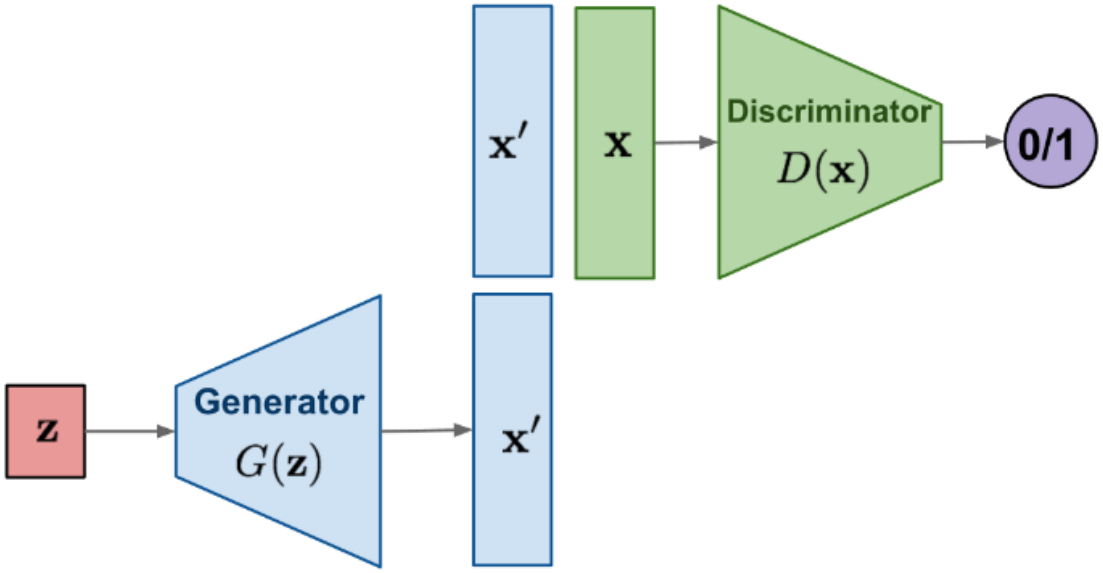


Generative Model Family

VAE: maximize variational lower bound

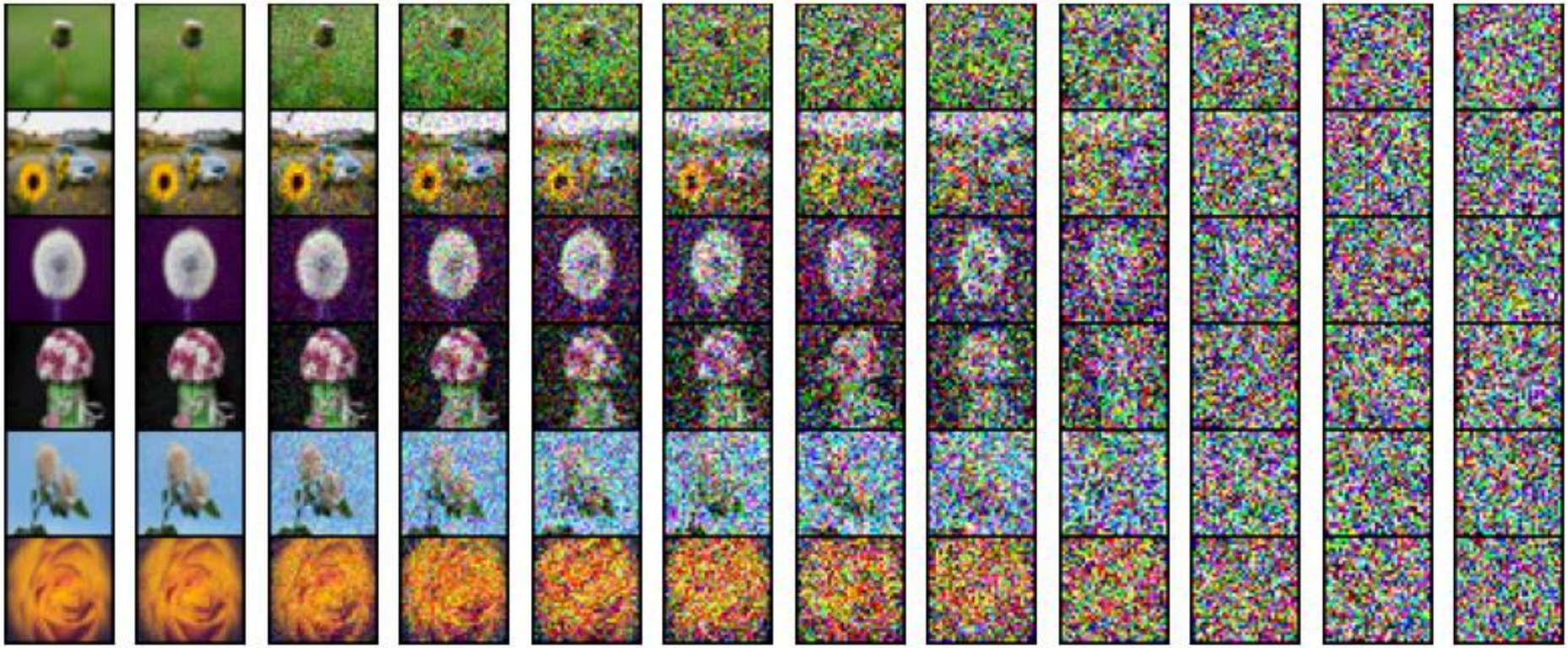


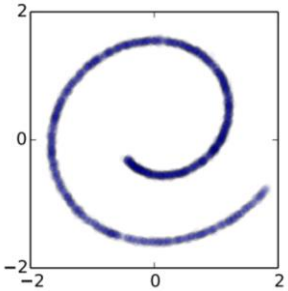
GAN: Adversarial training

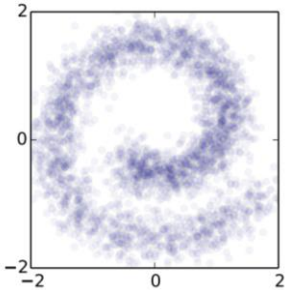


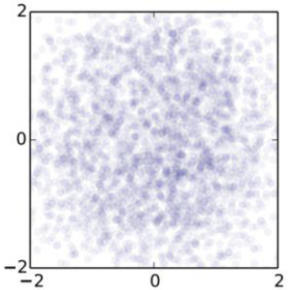
Diffusion models: Gradually add Gaussian noise and then reverse

What is Diffusion?



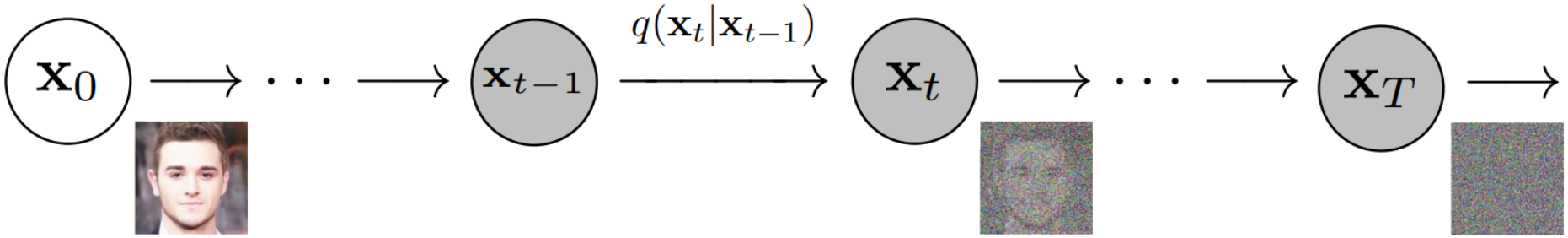




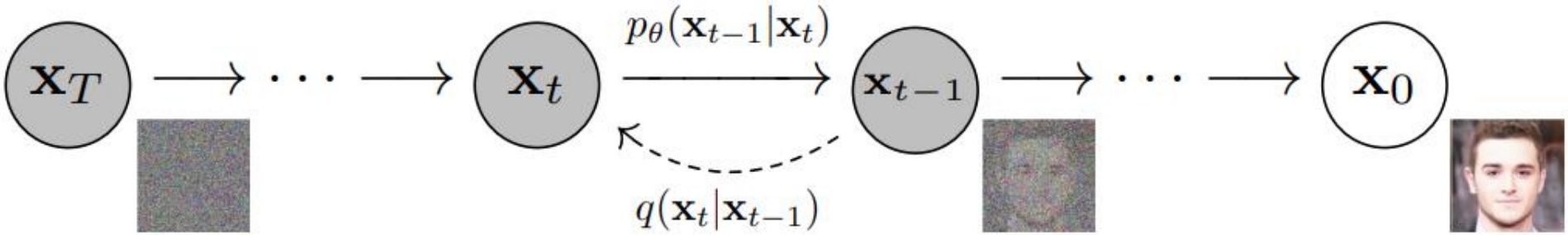


Diffusion Model Intro

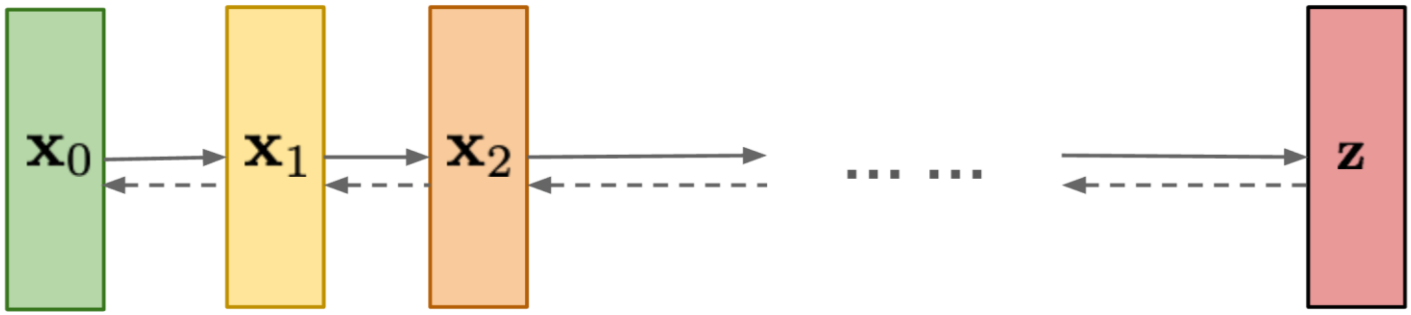
- **Forward Process: slightly add Gaussian noise for each step**
  - Until signal is destroyed

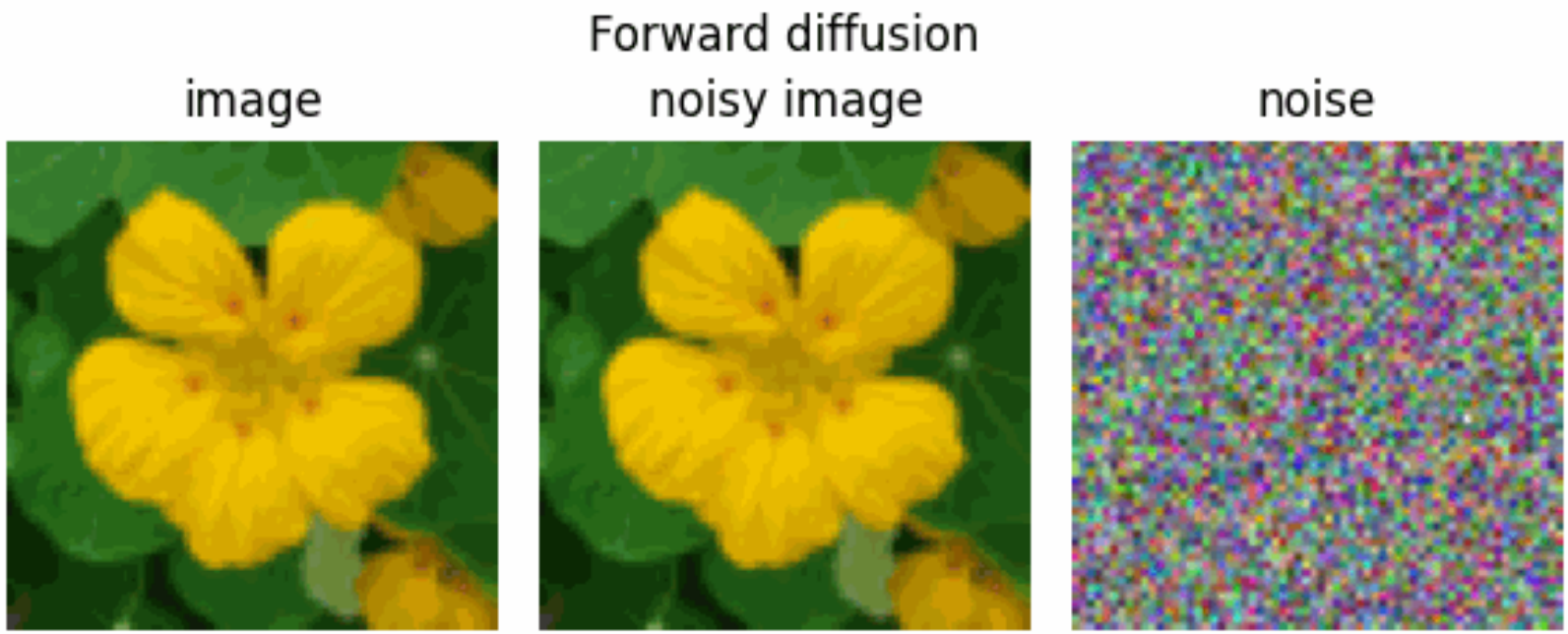


- **Reverse Process: learn how to slightly remove the noise for each step**



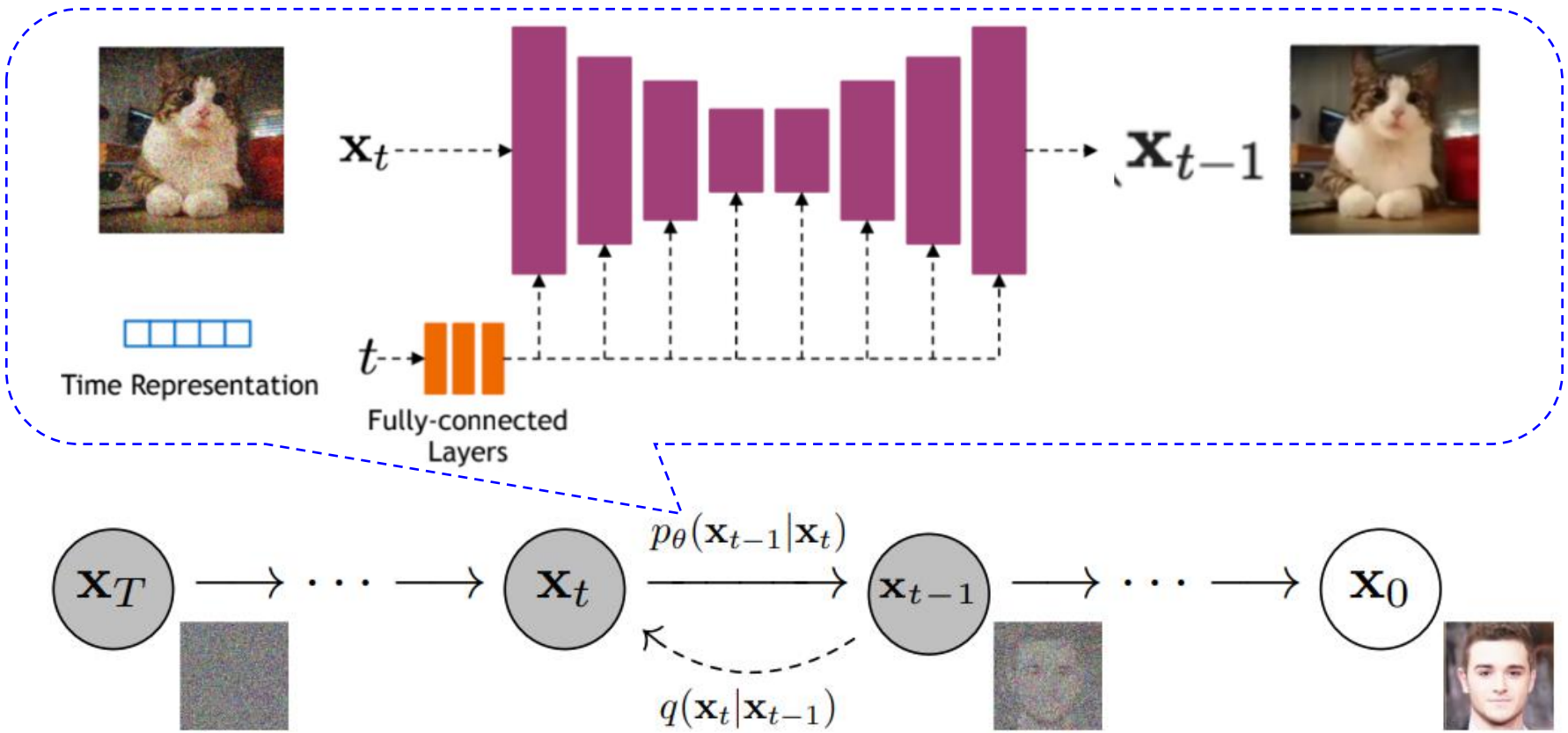
Diffusion Model Intro





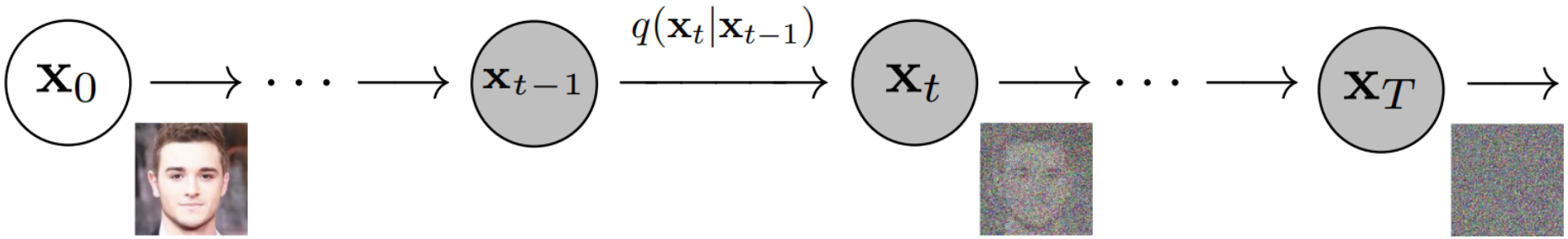
Diffusion Model Intro

- **Goal of model training**
  - Given an **arbitrary time step t** and the corresponding **sample**,
  - Learn a function that can produce a **slightly denoised version** of the given sample



Forward Process

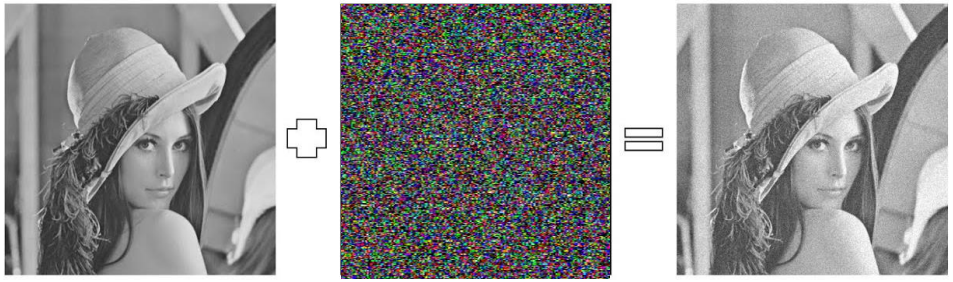
- **Training data:** forward process



- How can we model adding Gaussian noise to a given sample?
- A straightforward way:

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sigma_t \epsilon$$

$$\epsilon \sim \mathcal{N}(0, I)$$

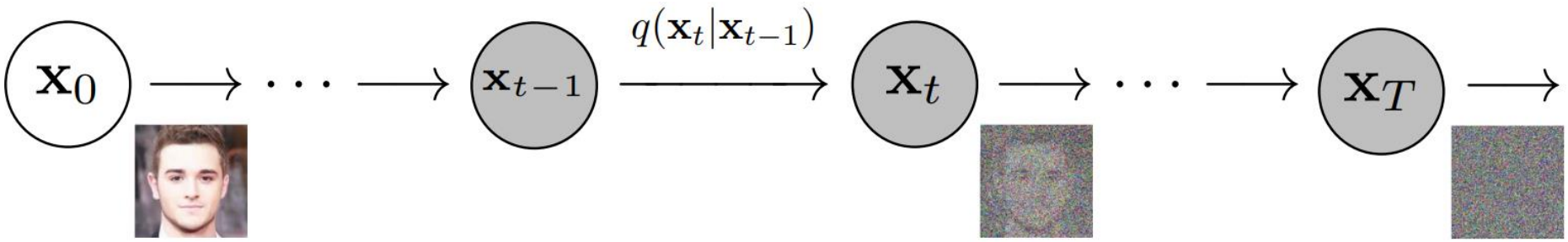


- But what happen if $t \to \infty$?

$\mathbf{x}_t \to \infty$. Not very useful

Forward Process

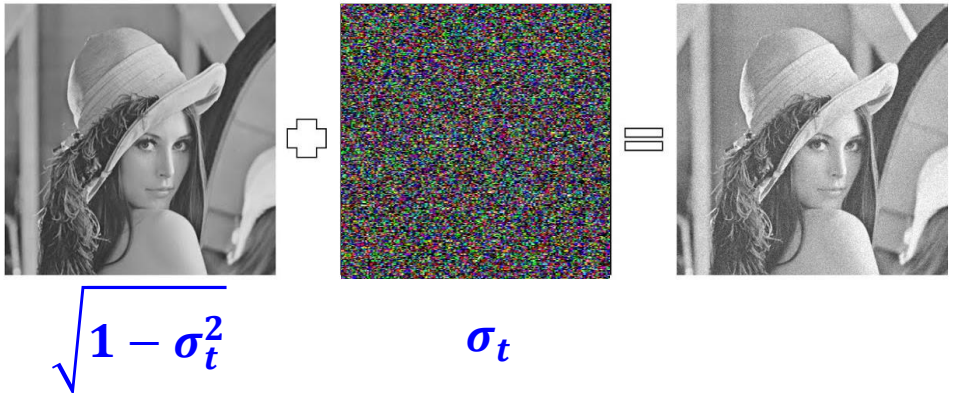
- **Training data:** forward process



- How can we model adding Gaussian noise to a given sample?
- What we actually do:

$$\mathbf{x}_t = \sqrt{1-\sigma_t^2} \cdot \mathbf{x}_{t-1} + \sigma_t \epsilon$$

$$\epsilon \sim \mathcal{N}(0, I)$$



- What happen if $t \to \infty$?

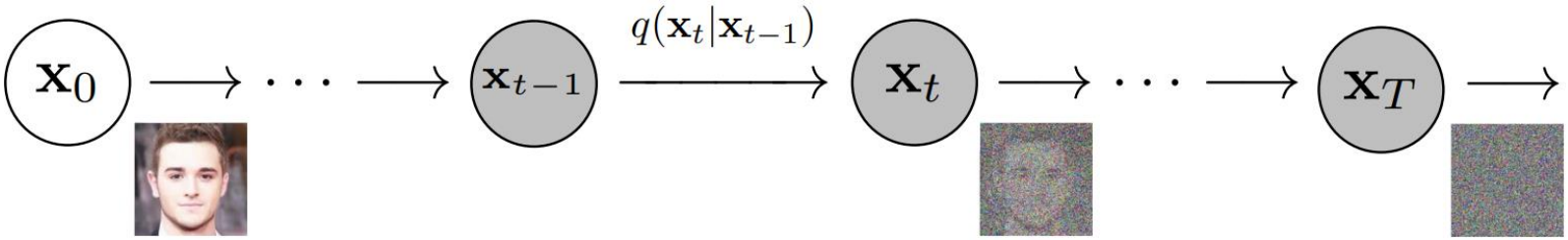
$p(\mathbf{x}_t) \to \mathcal{N}(0, I)$, the variance stays at 1.

Forward Process

*note: Gaussian distribution

$$\mathcal{N}(\mathbf{x} \mid \mu, \Sigma)$$

- **Training data:** forward process



- Make a generative-style formulation using Gaussian distribution

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}, \beta_t I\right) \qquad (\beta_t = \sigma_t^2)$$

  - This formulation remains the variance 1
  - β_t is linearly spaced from 10^{-4} to 0.02, with $T = 1000$.
  - β_t is pre-defined and fixed

*note:

$$Var(ax) = a^2 Var(x)$$

$$Var(x+y) = Var(x) + Var(y)$$

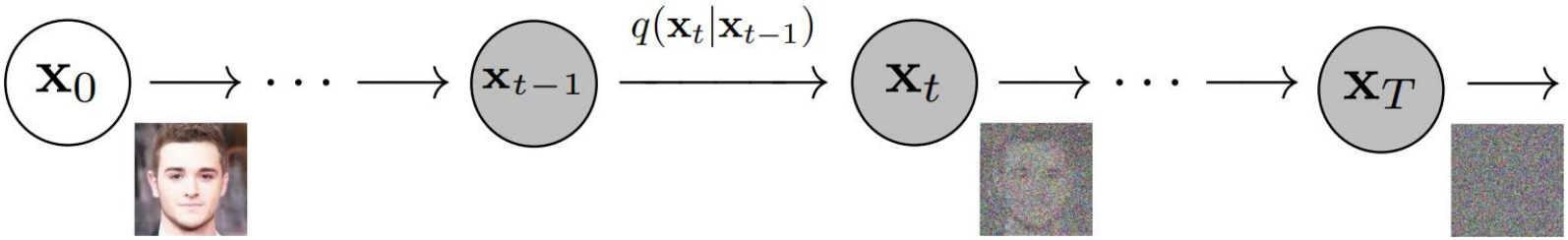
$$\hookrightarrow Var(\sqrt{1-\beta_t}x_t + \beta_t) = Var(\sqrt{1-\beta_t}^2 + \beta_t)$$

*note: re-parameterization

- Instead of sampling $z^{(i,l)} \sim \mathcal{N}\left(\mu(x^{(i)}), \Sigma(x^{(i)})\right)$
- We can derive $z^{(i,l)}$ in the same way by
  1. First sampling $\epsilon^{(l)} \sim \mathcal{N}(0, \mathrm{I})$
  2. Then computing
     $$z^{(i,l)} = \mu(x^{(i)}) + \sigma(x^{(i)}) \odot \epsilon^{(l)}$$
     - $\odot$: element-wise multiplication

Forward Process

- **Training data:** forward process



- How about jumping from x_0 to directly x_t ?

$$\mathbf{x}_t = \sqrt{1-\beta_t}\cdot\mathbf{x}_{t-1} + \sqrt{\beta_t}\epsilon,\ \ \epsilon\sim\mathcal{N}(0,I)$$

$$\alpha_t = 1-\beta_t, \quad \text{then,}\ \ \mathbf{x}_t = \sqrt{\alpha_t}\cdot\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\epsilon$$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} && \text{;where } \boldsymbol{\epsilon}_{t-1}, \boldsymbol{\epsilon}_{t-2}, \cdots \sim \mathcal{N}(\mathbf{0},\mathbf{I}) \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\bar{\boldsymbol{\epsilon}}_{t-2} && \text{;where } \bar{\boldsymbol{\epsilon}}_{t-2} \text{ merges two Gaussians (*).} \\
&= \cdots \\
&= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon} && \text{;where } \overline{\alpha_t} = \Pi_{s=1}^{t}\alpha_s
\end{aligned}$$

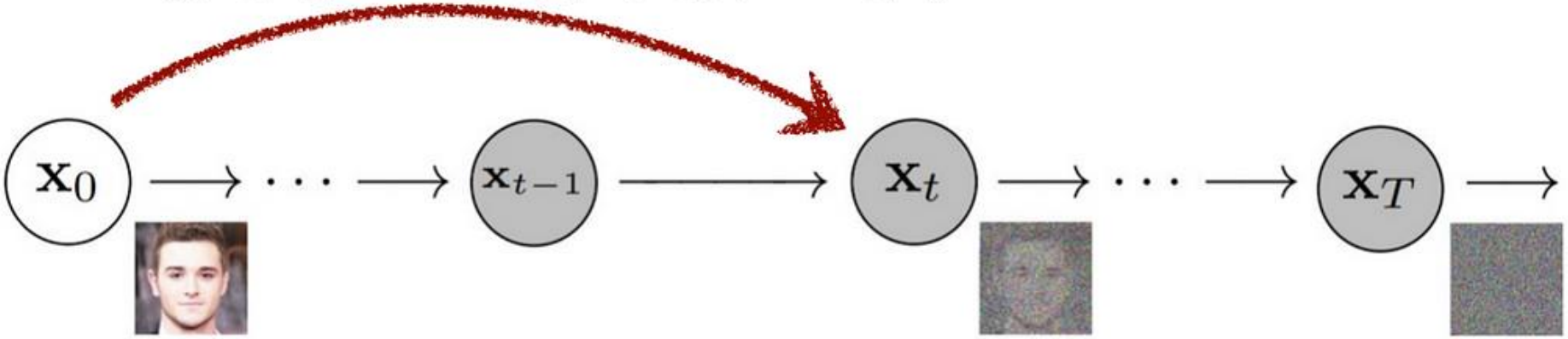
$$\boxed{q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})}$$

(*) Recall that when we merge two Gaussians with different variance, $\mathcal{N}(\mathbf{0}, \sigma_1^2\mathbf{I})$ and $\mathcal{N}(\mathbf{0}, \sigma_2^2\mathbf{I})$, the new distribution is $\mathcal{N}(\mathbf{0}, (\sigma_1^2+\sigma_2^2)\mathbf{I})$. Here the merged standard deviation is $\sqrt{(1-\alpha_t)+\alpha_t(1-\alpha_{t-1})} = \sqrt{1-\alpha_t\alpha_{t-1}}$.

Forward Process

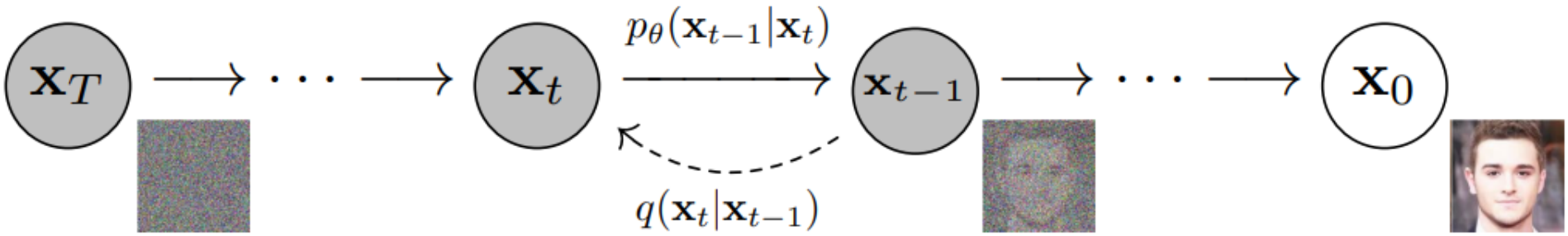
- **Training data:** forward process

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$



Reverse Process

Idea



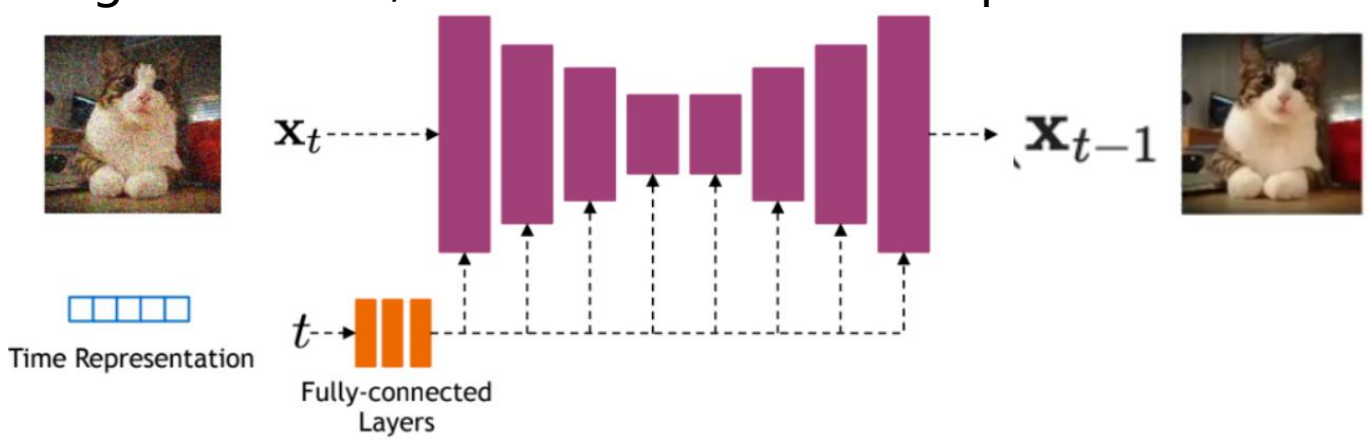
- Given a sample at t, produce its denoised version using a neural network

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t), \sigma_t^2 I)$$

isotropic covariance matrix

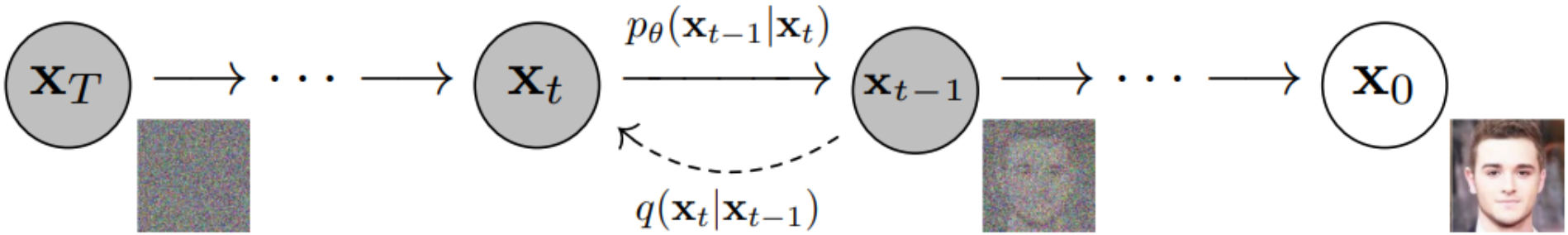
*note: Instead of training T networks, a neural net takes t as input

$\mu_\theta(x_t) = \mu_\theta(x_t, t)$



Reverse Process

- **Training objective: MLE (== minimizing negative log-likelihood)**



$$\arg\min_\theta \mathbb{E}_{\mathbf{x}_0}[-\log p_\theta(\mathbf{x}_0)] = \mathbb{E}_{\mathbf{x}_0}\Big[-\log \underbrace{\int p_\theta(\mathbf{x}_{0:T})\, d\mathbf{x}_{1:T}}_{\text{intractable}}\Big]$$

$$= \mathbb{E}_{\mathbf{x}_0}\left[-\log \int_{\mathbf{x}_{1:T}} \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} q(\mathbf{x}_{1:T}|\mathbf{x}_0)\right]$$

(Multiply & divide with the same term)

$$= \mathbb{E}_{\mathbf{x}_0}\left[-\log \mathbb{E}_q\left[\frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]\right]$$

*Expectation?

$E[f(X)] = \sum_{x_i \in X} f(x_i) \cdot P(x_i)$

$$\leq \mathbb{E}_{\mathbf{x}_0, q}\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]$$

*Jensen's inequality:

When $\varphi(x)$ is concave

$\varphi(\mathrm{E}[X]) \geq \mathrm{E}[\varphi(X)]$

Reverse Process

- **MLE modeling**

*note:

$$-\sum_{t>1}\log\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \log q(\mathbf{x}_T|\mathbf{x}_0) - \log q(\mathbf{x}_1|\mathbf{x}_0)$$

$$\mathbb{E}\left[-\log\frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}\left[-\log\prod_{t=1}^{T}\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)p(\mathbf{x}_T)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right]$$

$$= \mathbb{E}\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right]$$

$$= \mathbb{E}\left[-\log p(\mathbf{x}_T) - \sum_{t>1}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log\frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}\left[-\log p(\mathbf{x}_T) - \sum_{t>1}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}\cdot\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log\frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right]$$

$$= \mathbb{E}\left[-\log\frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t>1}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

*note: using the Markov property and the Bayes rule:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = q(\mathbf{x}_t|\mathbf{x}_{t-1},\mathbf{x}_0) = \frac{q(\mathbf{x}_t,\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}\frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}$$

Reverse Process

MLE modeling

$$\mathbb{E}\left[-\log\frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)}+\sum_{t>1}-\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$=\mathbb{E}\left[\underbrace{D_{KL}\big(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T)\big)}_{\text{no learnable components}}+\sum_{t>1}D_{KL}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)||\,p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$=\mathbb{E}\left[L_T+\sum_{t=2}^{T}L_{t-1}+L_0\right]$$

Reverse Process

- $L_{t-1} = D_{KL}\big(q(\mathbf{x}_{t-1}||\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)$
  - $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2 I\ )$
  - $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t I\ \big)$

    where $\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t$ and $\tilde{\beta}_t := \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$
- $L_{t-1} = \mathbb{E}_q\left[\boxed{\frac{1}{2\sigma_t^2}||\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)||^2}\right] + C$

*note: In general (K: dimensionality of variable)

$$D_{KL}[\mathcal{N}(\mu_0, \Sigma_0)||\mathcal{N}(\mu_1, \Sigma_1)]$$
$$= \frac{1}{2}\left(\mathrm{tr}(\Sigma_1^{-1}\Sigma_0) + \boxed{(\mu_1 - \mu_0)^\top \Sigma_1^{-1}(\mu_1 - \mu_0)} - K + \mathrm{In}\left(\frac{\det \Sigma_1}{\det \Sigma_0}\right)\right)$$

* tr: trace det: determinant

Reverse Process

*note: $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\cdot\mathbf{x}_{t-1}, \beta_t I)$

$q(\mathbf{x}_{t-1}|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_{t-1}; \sqrt{\bar{\alpha}_{t-1}}\cdot\mathbf{x}_0, (1-\bar{\alpha}_{t-1})I\right)$

$$p(\mathbf{x}) = \frac{1}{\sqrt{2\pi}^D|\Sigma|^{\frac{1}{2}}}\boxed{\exp\left(-\frac{1}{2}(\mathbf{x}-\mu)^\top\Sigma^{-1}(\mathbf{x}-\mu)\right)}$$

$\alpha_t = 1-\beta_t$, $\overline{\alpha_t} = \Pi_{s=1}^{t}\alpha_s$

*note:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0) = q(\mathbf{x}_t|\mathbf{x}_{t-1},\mathbf{x}_0)\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

$$\boxed{\propto \exp\left(-\frac{1}{2}\left(\frac{(\mathbf{x}_t-\sqrt{\alpha_t}\mathbf{x}_{t-1})^2}{\beta_t}+\frac{(\mathbf{x}_{t-1}-\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0)^2}{1-\bar{\alpha}_{t-1}}-\frac{(\mathbf{x}_t-\sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{1-\bar{\alpha}_t}\right)\right)}$$

$$= \exp\left(-\frac{1}{2}\left(\frac{\mathbf{x}_t^2-2\sqrt{\alpha_t}\mathbf{x}_t\mathbf{x}_{t-1}+\alpha_t\mathbf{x}_{t-1}^2}{\beta_t}+\frac{\mathbf{x}_{t-1}^2-2\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0\mathbf{x}_{t-1}+\bar{\alpha}_{t-1}\mathbf{x}_0^2}{1-\bar{\alpha}_{t-1}}-\frac{(\mathbf{x}_t-\sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{1-\bar{\alpha}_t}\right)\right)$$

$$= \exp\left(-\frac{1}{2}\left(\left(\frac{\alpha_t}{\beta_t}+\frac{1}{1-\bar{\alpha}_{t-1}}\right)\mathbf{x}_{t-1}^2-\left(\frac{2\sqrt{\alpha_t}}{\beta_t}\mathbf{x}_t+\frac{2\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}\mathbf{x}_0\right)\mathbf{x}_{t-1}+C(\mathbf{x}_t,\mathbf{x}_0)\right)\right)$$

$C(\mathbf{x}_t,\mathbf{x}_0)$ is some function not involving $\mathbf{x}_{t-1}$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t,\mathbf{x}_0), \tilde{\beta}_t I\right)$$

where $\tilde{\beta}_t = 1/\left(\frac{\alpha_t}{\beta_t}+\frac{1}{1-\bar{\alpha}_{t-1}}\right) = 1/\left(\frac{\alpha_t-\bar{\alpha}_t+\beta_t}{\beta_t(1-\bar{\alpha}_{t-1})}\right) = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\cdot\beta_t$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0) = \left(\frac{\sqrt{\alpha_t}}{\beta_t}\mathbf{x}_t+\frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}\mathbf{x}_0\right)/\left(\frac{\alpha_t}{\beta_t}+\frac{1}{1-\bar{\alpha}_{t-1}}\right)$$

$$= \left(\frac{\sqrt{\alpha_t}}{\beta_t}\mathbf{x}_t+\frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}\mathbf{x}_0\right)\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\cdot\beta_t$$

$$= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t+\frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0$$

Reverse Process

$$L_{t-1} = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}\|\underbrace{\boxed{\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0)}}_{\text{ground truth}} - \underbrace{\boxed{\mu_\theta(\mathbf{x}_t, t)}}_{\text{network prediction}}\|^2\right] + C$$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t$$

$$\mathbf{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_t)$$

$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_t\right)$$

*note: $\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$;where $\boldsymbol{\epsilon}_{t-1}, \boldsymbol{\epsilon}_{t-2}, \cdots \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

$= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\bar{\boldsymbol{\epsilon}}_{t-2}$;where $\bar{\boldsymbol{\epsilon}}_{t-2}$ merges two Gaussians (*).

$= \dots$

$\boxed{= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}}$;where $\overline{\alpha_t} = \Pi_{s=1}^t \alpha_s$, $\alpha_t = 1 - \beta_t$

$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$

(*) Recall that when we merge two Gaussians with different variance, $\mathcal{N}(\mathbf{0}, \sigma_1^2\mathbf{I})$ and $\mathcal{N}(\mathbf{0}, \sigma_2^2\mathbf{I})$, the new distribution is $\mathcal{N}(\mathbf{0}, (\sigma_1^2 + \sigma_2^2)\mathbf{I})$. Here the merged standard deviation is $\sqrt{(1-\alpha_t) + \alpha_t(1-\alpha_{t-1})} = \sqrt{1-\alpha_t\alpha_{t-1}}$.

Reverse Process

$$L_{t-1} = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}\|\underbrace{\boxed{\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0)}}_{\text{ground truth}} - \underbrace{\boxed{\mu_\theta(\mathbf{x}_t, t)}}_{\text{network prediction}}\|^2\right] + C$$

*note: $\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$

$= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\bar{\boldsymbol{\epsilon}}_{t-2}$

$= \dots$

$\boxed{= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}}$

$$\tilde{\boldsymbol{\mu}}_t = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_t\right)$$

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right)$$

$$L_{t-1} = \mathbb{E}_{\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\frac{1}{2\sigma_t^2}\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t, t)\|^2\right]$$

$$= \mathbb{E}_{\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\frac{1}{2\sigma_t^2}\|\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_t\right) - \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right)\|^2\right]$$

$$= \mathbb{E}_{\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t)\,\sigma_t^2}\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2\right]$$

$$= \mathbb{E}_{\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t)\,\sigma_t^2}\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_t, t)\|^2\right]$$

Reverse Process

Final objective

$$L_{t-1} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0},\mathbf{I})} \left[\frac{\beta_t^2}{2\sigma_t^2 (1-\beta_t)(1-\bar{\alpha}_t)} ||\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}\, \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, t)||^2 \right]$$

$$L_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0},\mathbf{I}), t \sim \mathcal{U}(1,T)} \left[||\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}\, \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, t)||^2 \right]$$

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

*note:

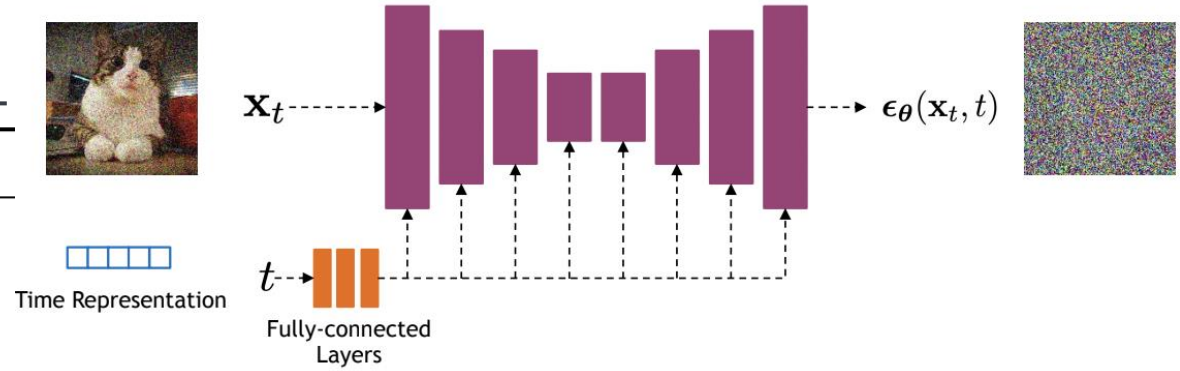
$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right)$$

$$\text{Thus } \mathbf{x}_{t-1} = \mathcal{N}(\mathbf{x}_{t-1}; \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right), \sigma_t^2 I)$$

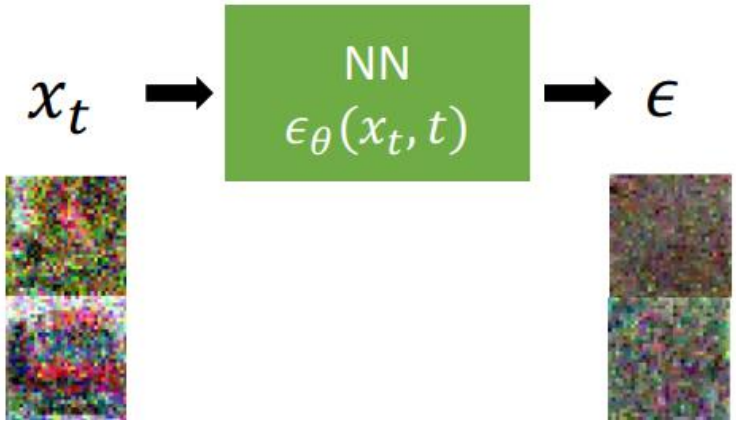
Reverse Process

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged



"How much noise is in this diffused example?"



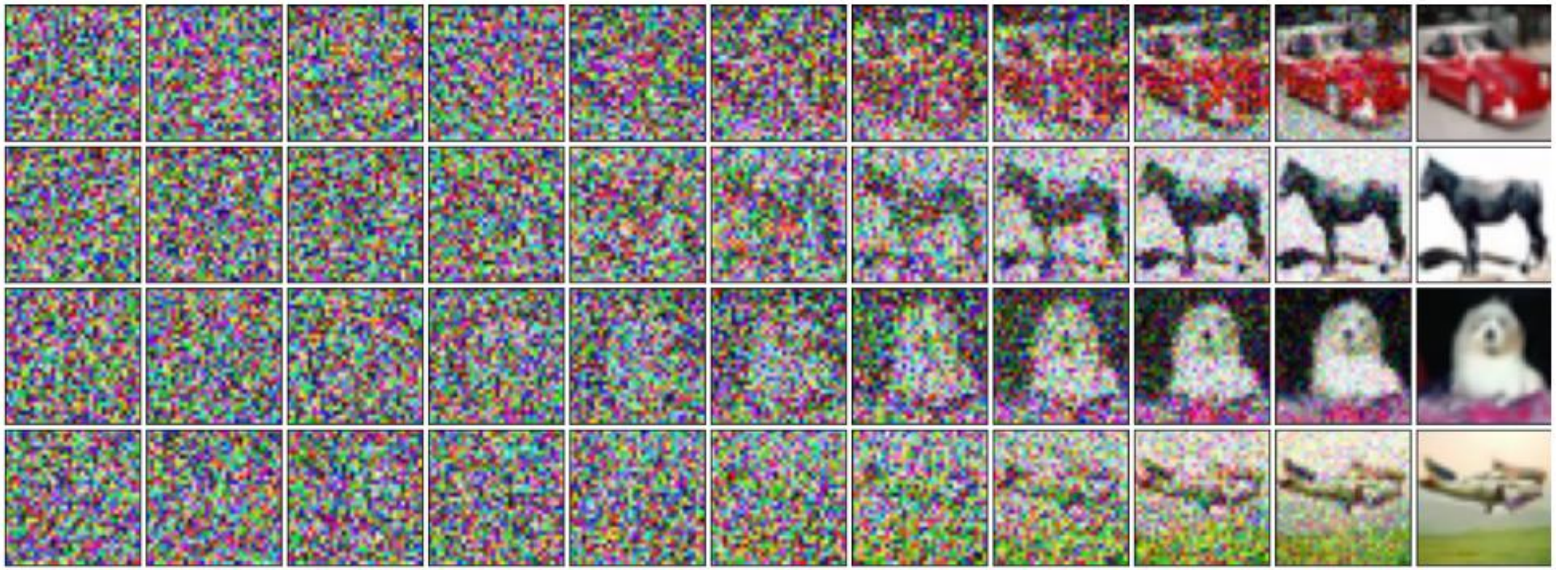
1. Given $\mathbf{x}_0$ (Data)
2. Compute $\mathbf{x}_t = \sqrt{\bar{a}_t}\mathbf{x}_0 + \sqrt{1-\bar{a}_t} \cdot \epsilon$
3. Train to predict $\boldsymbol{\epsilon}$ given $\mathbf{x}_t$ by minimizing

$$||\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\right)||^2$$

- This is more easier than learning $\mathbf{x}_{t-1} - \mathbf{x}_t$.
- Very Similar to Denoising Autoencoders (Vincent et al., 2008) and other score-based models

Results

- **Sampling**



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Thank You